

# PRISCILLA MALOMBE NZULA

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

## PROFESSIONAL SUMMARY

Credit Scoring Analyst and data science professional with 5+ years in banking industry, including 2+ years building production ML models. Currently designs and validates credit scorecards at National Bank of Kenya, monitoring portfolio risk using GINI, KS and PSI. Managed a KES 450 million loan portfolio across 30,000 clients at Equity Bank, then moved into independent data science practice, delivering production-ready models, live dashboards, and automated pipelines across manufacturing, HR, and retail domains. Skilled in Python, SQL and cloud deployment.

## TECHNICAL SKILLS

**Credit Risk & Scoring:** Scorecard Development, Model Validation, Back-testing, GINI/KS/PSI Monitoring, Risk-Based Pricing, IFRS9, Basel, CBK Compliance.

**Programming and Libraries:** Python, NumPy, Pandas, Scikit-learn, TensorFlow, Keras, XG Boost, Joblib.

**Machine Learning:** Predictive Modeling, Random Forest, Isolation Forest, Feature Engineering, SMOTE, Deep Learning, Transfer Learning, NLP.

**Database and SQL:** MySQL, Supabase, Querying, ETL Processes, Database Design, Stored Procedures.

**Data Visualization:** Streamlit, Plotly, Matplotlib, Seaborn, Chart.js, Power BI, Excel.

**Deployment and Cloud:** GitHub Actions, Streamlit Community Cloud, Netlify, Fast API, CI/CD Pipelines.

**Tools and Platforms:** Jupyter Notebook, VS Code, Google Colab, Git, GitHub.

## WORK EXPERIENCE

### Credit Scoring Analyst

May 2026 – Present

National Bank of Kenya | Nairobi, Kenya (Contract)

- Design, build, and validate application and behavioral credit scorecards using Python and machine learning, supporting data-driven lending decisions across the portfolio.
- Monitor model performance using GINI, KS, and PSI, recommending recalibrations to maintain scorecard accuracy over time.
- Extract, clean, and validate large datasets from SQL and Oracle databases to support model development and credit risk assessment.
- Support risk-based pricing and segmentation strategies in collaboration with IT, data, and business teams.
- Ensure model and reporting practices align with regulatory requirements, including CBK, IFRS9, and Basel.
- Build dashboards and reports tracking score performance and portfolio risk, supporting automation of the credit decisioning process.

### Data Scientist (Freelance)

September 2024 – April 2026

Independent Projects | Nairobi, Kenya

- Delivered a manufacturing analytics project for turbofan engine maintenance, enabling the business to identify critical engines and schedule interventions in advance rather than reacting to unexpected failures.
- Built an HR analytics solution that gave a people team early visibility into which employees were most likely to leave, surfacing the departments and workload patterns driving the highest turnover.
- Produced a retail analytics report for a fashion business that clarified which customer segments and product categories were driving the majority of revenue, informing inventory and marketing priorities.
- Developed a telecoms churn model that identified customers most likely to cancel their contracts, giving the retention team a prioritized list to act on before attrition occurred.
- Designed and implemented two production database systems covering e-commerce and library operations, replacing ad hoc data handling with structured schemas and automated data integrity checks.

### Relationship Officer, Credit Monitoring and Collections

November 2021 – August 2024

Equity Bank Limited | Nairobi, Kenya

- Analyzed financial data and loan repayment patterns for a portfolio of 30,000 clients across Corporate, SME, and Retail segments valued at KES 450 million, identifying trends and default risks.
- Built and maintained Excel dashboards to monitor portfolio performance, track overdue payments, and present key risk indicators to management.

- Applied data-driven insights from customer payment histories and financial statements to support account restructuring and creditworthiness assessments, contributing to a 30% reduction in monthly provisioning and account slippage.
- Extracted and synthesized client data from core banking systems (FinOne, Finacle) to ensure accurate reporting and compliance with credit monitoring protocols.

### Relationship Officer, Cash

August 2019 – October 2021

Equity Bank Limited | Nairobi, Kenya

- Processed and reconciled high-volume client transactions, ensuring accuracy and compliance with financial regulations.
- Monitored client account activity to identify irregular patterns and flag potentially fraudulent transactions for investigation.
- Compiled daily transaction reports to track branch performance metrics, providing data that supported management decision-making.

## PROJECTS

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### Turbofan Engine Predictive Maintenance | Python, Random Forest, MySQL, Supabase, Streamlit, GitHub Actions

[Link](#)

- Trained a Random Forest regressor on 20,631 NASA CMAPSS sensor readings to predict engine Remaining Useful Life, achieving  $R^2 = 0.969$  and RMSE = 7.2 cycles. Engineered rolling average and standard deviation features over a 20-cycle window across four key sensors to capture degradation trends.
- Connected the model to a Supabase cloud database, built a live Streamlit dashboard showing fleet health across 100 engines, and automated daily data refresh and maintenance alerts via a GitHub Actions cron workflow.

### HR Analytics: Employee Attrition Prediction | Python, Random Forest, XGBoost, FastAPI, SMOTE

[Link](#)

- Trained and compared Random Forest and XGBoost classifiers on 29,524 employee records, applying SMOTE to correct class imbalance. The best model achieved 0.87 ROC-AUC and caught 75% of at-risk employees before they resigned.
- Found a 33% company-wide attrition rate concentrated in Finance, Sales, and Operations, with number of projects, monthly hours, and satisfaction score as the top three predictors. Deployed the model as a FastAPI REST endpoint for real-time scoring.

### StyleSense: Retail Sales and Customer Analytics | Python, Pandas, RFM Segmentation, Plotly

[Link](#)

- Segmented 12 months of retail transaction data using RFM analysis, classifying customers into Champions, Loyal, At-Risk, and Lost groups. Found that the top 20% of customers generated over 80% of total revenue.
- Forecasted seasonal demand patterns and identified the highest-margin product categories, producing a ranked list of restocking priorities and markdown recommendations backed by transaction-level data.

## EDUCATION

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### BSc Mathematics and Computer Science

September 2015 – October 2019

Multimedia University of Kenya

## CERTIFICATIONS

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Data Science Certification, Moringa School

August 2023 – June 2024

Data Protection Certification, CIPIT Strathmore University

October 2024